

Study Guide and Glossary

A revision list for explaining the platform to an industry jury: what it does, how log formats differ, what the prototype proves, and what must be built before production.

How to use this guide. Start with the six facts, rehearse the demo route, then review terms tagged JURY / KNOW. Definitions separate standard concepts from current LogProof behavior.

FIRST: KNOW	THEN: EXPLAIN
Problem and user	Formats and field mapping
Live demo path	Evidence and parser assurance
Verified limits	Scale, security, deployment
Dataset caveats	Adoption and next milestones

One-sentence product description

LogProof is a pre-SIEM assurance prototype that keeps received log bytes inspectable, shows how fields were interpreted, flags a firewall format change, and replays a candidate parser before an operator approves it.

Claim discipline: say “the prototype demonstrates” for implemented behavior; say “the design proposes” for future production capabilities.

SIH slide-defense checklist

Slide	Study / verify
Problem	Name the affected user and the perimeter-log preprocessing gap; use the exact SIH problem statement.
Solution	Explain the pre-SIEM boundary and the narrow evidence / replay difference.
Technical approach	Show receive → preserve → parse → map → review, supported formats, and one traceable field.
Feasibility	Separate built, tested, and proposed. State prototype limits; do not equate a 30-hour demo with ministry readiness.
Impact	Propose measurable pilot outcomes such as onboarding time, mapping coverage, drift detection, and rollback recovery; label targets as targets.
Research / team	Cite primary standards and dataset caveats. Check problem ID, theme/team fields, names, and any demo/video placeholder before presenting.

1. What to study first

Memorize these six points before revising the glossary. They give you a clear path through a short pitch and demo.

Problem and user	Security and platform teams receive differently structured logs. Source-specific parsing slows onboarding, investigation, and safe parser changes. The primary user is the engineer or security operator maintaining sources.
Product boundary	LogProof sits before a SIEM or data lake as a preprocessing and verification layer. It is not a SIEM, threat detector, or replacement for collection pipelines.
Distinctive proof	Raw receipt → field map → firewall shape drift → same-corpus replay → human approval → rollback target. Universal preprocessing has prior art; focus differentiation on reviewable provenance and parser-change assurance.
Working scope	Ten sample source families span eight raw formats: JSON, CSV, RFC 3164-style Syslog, RFC 5424 Syslog, CEF, NGINX combined, Windows Event XML, and LEEF. Parsers are representative, not universal. Only the firewall parser has full replay and promotion.
Numbers and limits	Single event up to 256 KB. NDJSON: 100 records, 2 MB total, 256 KB per record. A sequential local run of 1,000 synthetic events measured 62.57 events/second; this is not production capacity.
Deployment answer	Local files + SQLite + one API service support the demo. A no-pull image archive was loaded on the same Docker engine. Separate ministry air-gap host, SIEM export, high availability, and ministry-scale load are unverified.

Platform flow



Invalid or incomplete records go to quarantine; raw bytes remain inspectable.

Raw bytes and normalized fields stay linked by receipt metadata. A digest can show whether stored bytes changed after receipt; it cannot identify the original sender.

Three-minute demo route

1	Open Live ingestion. Show source profiles and import one event.
2	Open Evidence. Select a normalized field and show its source key/byte span and receipt digest.
3	Trigger the firewall shape change. Show the changed path in Drift Watch; inspect the raw event.
4	Run firewall replay. Explain same corpus, candidate differences, gates, approval, and rollback target.
5	Show a malformed event in Quarantine. Close with storage, security, and scale limits.

Glossary / Log and event basics

Log / event record JURY / KNOW	A time-associated record of an action or state change. “Log” often means the stored message; “event” emphasizes what happened.
Source / producer JURY / KNOW	The device, OS, service, or application that emitted an event. A source label is not proof of sender identity.
Collector DEMO	A component that receives or reads logs and forwards them. LogProof accepts HTTP API submissions and simulator events; it has no network listener.
Structured / unstructured JURY / KNOW	Structured records have explicit keys/elements; unstructured records require pattern or delimiter parsing. Both can change shape.
JSON JURY / KNOW	Nested text objects and arrays with typed values. Parsers must handle missing keys, changed types, and extra fields.
CSV DEMO	Rows and columns separated by a delimiter. Headers, quoting, escaping, encoding, and data types affect parsing. Demo accepts a header and one event.
XML JURY / KNOW	A tagged tree. Namespaces, parent/child structure, attributes, and optional elements matter; avoid parsing XML by splitting strings.
NDJSON / JSON Lines DEMO	One UTF-8 JSON value per line. The demo uses one source/raw or source/raw_base64 envelope per line for mixed-source batches.
Syslog JURY / KNOW	A family of message formats and transports common in network and Unix-like systems. Message format and transport are separate concerns.
RFC 5424 STANDARD	Standardized Syslog message: PRI VERSION TIMESTAMP HOSTNAME APP-NAME PROCID MSGID STRUCTURED-DATA, then optional MSG.
PRI / facility STANDARD	Syslog PRI combines facility and severity: $PRI = facility \times 8 + severity$. Facility identifies the generating subsystem; it is not severity.
Syslog severity STANDARD	RFC 5424 levels: 0 emergency, 1 alert, 2 critical, 3 error, 4 warning, 5 notice, 6 informational, 7 debug. Lower number means more severe.
RFC 3164 / BSD Syslog STANDARD	Legacy observed message style, not an Internet Standards Track format. Its timestamp can omit year and timezone.
CEF STANDARD	Common Event Format associated with ArcSight. Prefix: CEF:version vendor product device-version signature-id name severity , then extension fields.
LEEF STANDARD	Log Event Extended Format associated with QRadar. Header identifies version/vendor/product/product version/event ID; attributes are key/value pairs. LEEF 2 supports alternate delimiters.
NGINX combined access DEMO	Text access record commonly containing client IP, timestamp, HTTP request, response status, size, referrer, and user-agent.
Windows Event XML DEMO	Event record with structured System and event data, often provider, event ID, time, channel, computer, and named values. Demo also has a separate simplified Windows JSON sample.
Firewall / network flow DEMO	Firewall events record rule/action and traffic fields; flow records summarize communication over a connection or interval, not packet contents.

Glossary / Normalization, schema, and time

Parsing JURY / KNOW	Converting source bytes/text into typed fields using a source-specific rule. A parser can fail or produce plausible but wrong values.
Normalization JURY / KNOW	Mapping source fields and values into shared names and meanings while retaining source context and uncertainty.
Canonical schema DEMO	The shared event model inside a product. LogProof uses a limited project schema; it is not complete OCSF.
OCSF STANDARD	Open Cybersecurity Schema Framework: extensible event schema, attribute dictionary, and taxonomy. It is a schema, not a collector or storage engine.
OpenTelemetry Logs Model STANDARD	Vendor-neutral log record fields include timestamps, severity, body, resource, instrumentation scope, attributes, and event name. Distinct from LogProof's schema.
Field map DEMO	Shows normalized field, source key, transform, confidence, and optional source-byte offsets.
Event time / observed time JURY / KNOW	Event time says when the source says it happened; observed time says when a processor received it. They differ with late delivery or clock skew.
UTC normalization DEMO	Representing parsed times in a common basis such as ISO 8601 UTC. Retain source bytes and mapping so conversion is reviewable.
Severity mapping JURY / KNOW	Mapping source-specific levels to a common scale. Preserve original value: Syslog 0–7 counts down in severity; OTel 1–24 increases.
Missing / null / coercion JURY / KNOW	Missing means absent; null is explicit. Coercion converts types (such as digits to integer) and should be observable and tested.

Glossary / Evidence, integrity, and parser assurance

Raw retention DEMO	Keep received bytes available for inspection. LogProof writes raw data before parsing; access, backup, retention, and secure deletion remain operational duties.
Lossless JURY / KNOW	Recover original bytes and meaning without alteration. Raw bytes are retained, but normalized fields alone do not preserve every source semantic.
Receipt ID DEMO	Identifier linking one submission to raw storage and processing metadata; does not prove authenticity.
SHA-256 DEMO	Digest that detects whether stored bytes match the value recorded at receipt. It does not authenticate the sender or trusted time.
Integrity / authenticity JURY / KNOW	Integrity checks for change relative to trusted evidence. Authenticity identifies who sent/created data. Current hash check only supports local byte-integrity.
Provenance / lineage DEMO	Metadata about event origin and transformations. LogProof aims to trace each normalized value to source material and parser version.
Byte span DEMO	Start/end positions of a field in raw bytes. Useful for trace-back; unavailable for some transformations and structured dataset fields.
Parser drift / baseline DEMO	Drift is a change in source field names, types, or nesting. Baseline is the expected shape; drift is not itself threat detection.
Quarantine DEMO	Hold malformed or incomplete records for inspection while retaining raw data. It does not automatically repair or replay them.
Golden corpus / replay DEMO	Fixed expected examples form a golden corpus. Replay applies active and candidate parsers to the same records for comparison.
Regression / promotion DEMO	Regression is a previously correct outcome made worse. Promotion activates a candidate after checks and human approval; demo approval name is not authenticated.
Rollback / signature JURY / KNOW	Rollback returns to the prior parser version. A checksum detects change only against a trusted value; a digital signature verifies a key holder signed bytes. Demo packs are not signed.

Glossary / Architecture, interfaces, and operations

API / endpoint DEMO	An interface and its method, path, request, response, and errors. Example: POST /api/ingest/batch. Backend is FastAPI.
Batch ingestion DEMO	Multiple envelopes per request; each NDJSON line is handled independently, allowing partial success.
Pagination DEMO	Bounded slices for large lists. Demo uses 20-item limit/offset pages; offset can be inefficient at very large scale.
SQLite / raw-file store DEMO	SQLite stores metadata; local files store original bytes. Separate writes can fail halfway; there is no cross-store transaction.
Container image JURY / KNOW	Packaged application and runtime. An image is not a running service, data backup, or proof of a separate air-gap deployment.
Air gap JURY / KNOW	Restricted network without ordinary external access. Strong proof requires transfer, verification, and offline start on a separate target host.
SIEM / SOAR / data lake JURY / KNOW	SIEM centralizes security analysis; SOAR coordinates response; a data lake stores data for broader analytics. LogProof currently exports to none of them.
Throughput / latency JURY / KNOW	Throughput is events per time. Latency is time per request/event; state workload and report p50/p95/p99, not only averages.
Microbenchmark DEMO	A narrow bounded measurement. LogProof's 1,000-event run is sequential and excludes HTTP/concurrency; 62.57 events/s is not production capacity.
Concurrency / backpressure NEXT / LIMIT	Concurrency runs work simultaneously. Backpressure slows/rejects producers when processing falls behind. Neither was benchmarked/implemented as a streaming queue.
Idempotency / delivery JURY / KNOW	Idempotency makes retries safe. Manual uploads create new receipts; avoid "exactly once" claims without defined delivery and deduplication semantics.
HA / SLO NEXT / LIMIT	High availability covers failure and recovery; an SLO is a measurable service target over time. The prototype has no production HA or SLO.

Glossary / Security, governance, dataset, and product terms

Authentication / authorization / RBAC NEXT / LIMIT	Authentication establishes identity; authorization decides permitted actions; RBAC assigns permissions through roles. None is implemented in the demo.
Tenant isolation NEXT / LIMIT	Separates each organization's data and actions. Hosted demo state is shared synthetic data, not a tenant-isolation design.
Data minimization / retention / residency JURY / KNOW	Collect only needed data; define how long it is stored and where. Raw logs can contain secrets or personal data.
Threat model NEXT / LIMIT	Lists assets, boundaries, attackers, threats, and mitigations. Needed before handling real sensitive logs.
Synthetic / sanitized data JURY / KNOW	Synthetic samples are crafted; sanitized data has sensitive details altered or removed. Sanitization can reduce useful context.
WitFoo Precinct6 dataset DEMO	Public card describes sanitized SOC-derived event data and incident graphs. The app fetches a bounded `signals/train` slice; it does not download the full dataset.
Publisher-derived label JURY / KNOW	WitFoo labels come from its automated incident correlation, not independent analyst-verified ground truth for every row. Call them publisher-provided machine-derived labels.
Representative data JURY / KNOW	Reflects the target environment's sources and distribution. Ten crafted examples or one public dataset cannot prove ministry-wide coverage.
Pre-SIEM assurance layer JURY / KNOW	Product position: verify interpretation and parser changes before downstream analytics. This is an intended role; no SIEM integration exists yet.
Differentiation / value hypothesis JURY / KNOW	Differentiate narrowly on field-level provenance, drift review, replay impact comparison, and safer parser promotion; validate value with source-onboarding time and operator feedback.
Logstash / Fluent Bit JURY / KNOW	Examples of established pipeline tools. Logstash uses input/filter/output stages; Fluent Bit is a lightweight collector with input/filter/output plugins and buffering. Compare exact deployment and parser capabilities.
OpenTelemetry Collector STANDARD	Vendor-neutral telemetry pipeline with receivers, processors, and exporters; supports logs, metrics, and traces. It is not the same thing as the OpenTelemetry log record data model.
Cribl Stream JURY / KNOW	Collects, processes, transforms, and routes event data through sources, pipelines, functions, and destinations. Its docs include preview and verification workflows; do not claim competitors lack testing.
Competitor comparison JURY / KNOW	Compare documented versions against specific needs: source coverage, format parsing, raw retention, field lineage, change replay, approval, rollback, air-gap, and outputs. Avoid blanket feature claims.
Pilot / license hypothesis NEXT / LIMIT	LogProof's proposed pricing shows a paid pilot and annual self-hosted tiers. Treat those figures as a proposal, not validated willingness-to-pay or an approved ministry procurement plan.

Jury questions to practise

Answer in this order: direct answer → demo proof → current limit → next validation step. Keep every claim within the evidence.

JURY QUESTION	ANSWER TO PRACTISE
What is different from a parser or SIEM?	We focus on field-level provenance and controlled parser changes upstream. The demo links fields to raw receipts, detects firewall shape drift, and replays a candidate before approval. It does not replace SIEM analytics.
Does the hash prove that the device sent the log?	No. SHA-256 checks stored bytes against a digest captured at receipt. Device identity needs authenticated transport or signed evidence with managed keys.
Are these production logs?	Most demo examples are synthetic. A bounded WitFoo slice is sanitized SOC-derived data with publisher-provided automated labels, not universal coverage or independent ground truth.
Can it handle every vendor and format?	No. It has representative parsers across eight raw formats and ten sample families. Every added source needs sample validation, mapping, tests, and drift review.
Why not map fully to OCSF?	OCSF is a strong interoperability target. The current internal schema is smaller; complete mapping, versioning, profile/extension choices, and conformance tests remain future work.
How do you know parser changes are safe?	The firewall example replays fixed golden examples and recent firewall events through active and candidate versions, reports differences and regressions, then gates approval.
Will it scale for the ministry?	Not proven. The measured result is a sequential local microbenchmark. Define source rates, bursts, retention, latency and availability first, then test concurrency, storage, recovery, and export.
Is it air-gap ready?	A no-pull archive was started on the same Docker engine. A separate isolated host is untested; the optional dataset fetch needs internet.
What protects sensitive logs?	The demo lacks authentication, RBAC, tenant isolation, and production retention/key management. Use synthetic data until controls and a threat model are implemented.
What is the next milestone?	Choose target sources and policies; add access control and robust persistence; test SIEM export and separate-host offline installation; then run a measured pilot with recovery and rollback gates.

Primary study references

Read the standards for format semantics and the repository for exact prototype behavior. Dataset details can change; verify live-card counts. Checked 25 September 2026.

RFC 5424 — The Syslog Protocol

RFC 3164 — The BSD Syslog Protocol

ArcSight CEF Implementation Standard

IBM QRadar — LEEF event components

OCSF — Open Cybersecurity Schema Framework

OpenTelemetry Logs Data Model

NDJSON specification

Microsoft Windows Event schema

NIST SP 800-92 — Log Management

WitFoo Precinct6 dataset card

Cribl Stream — Work with Data

Elastic Logstash — How it works

Fluent Bit — Official Manual

OpenTelemetry Collector — Components